

1. Problem Statement

Crime data across State Police districts is fragmented across paper FIRs, spreadsheets, and disconnected station records. Command staff have no single, live view of crime trends and hotspots to direct patrols and resources; investigating officers have no unified workspace that surfaces a case's linked suspects, victims, evidence and timeline in one place. The result is slower investigations, delayed pattern recognition across similar cases, and no accessible, government-grade portal that both decision-makers and field officers can rely on.

2. Key Features & Functionalities

- **Statewide Dashboard** — live registered-case totals, crime-category counts, districts covered, and the 112 emergency helpline, at a glance.
- **Crime Count & Hotspots** — aggregated crime statistics alongside an embedded 3D / spatiotemporal crime map for geographic hotspot analysis.
- **Cases → District-wise Drilldown** — browse by crime category (Theft, Assault, Fraud, Burglary); see per-district counts with 5-year trend sparklines and clearance-rate indicators, fully searchable and filterable.
- **Investigation Workspace** — a victim-centred detective evidence board linking suspects, witnesses, vehicles, phones, bank accounts and evidence via labelled relationship threads, plus a timeline, evidence locker, AI-generated modus-operandi & risk insights, and related-case matching.
- **Digital Case-File Booklet** — a 12-page flip-through digital FIR covering cover, incident, crime scene, victim, suspects, witnesses, evidence, notes, AI analysis, similar cases, final report, and a dedicated evidence-board page.
- **Accessibility-first design** — WCAG 2.1 AA, adjustable font size, high-contrast mode, skip-navigation and full keyboard support.
- **Government design language** — a dignified navy/white portal aesthetic modelled on official State Police branding, responsive on mobile.
- **Officer authentication** — sign-in gate scaffolded on Zoho Catalyst Authentication, ready to enforce role-based access.
- **Live backend, not a mock-up** — a genuine Zoho Catalyst Data Store modelled on the Karnataka Police FIR schema, served through Catalyst Serverless Functions.

3. Technology Stack

Frontend	Next.js 15 (App Router, static export) · React 19 · TypeScript · Tailwind CSS · Framer Motion · D3-force · Lucide Icons
Cloud / Backend	Zoho Catalyst · Web Client Hosting (static frontend) · Serverless Functions (Node.js / Express REST API over ZCQL) · Data Store (relational FIR database) · Authentication (officer sign-in, scaffolded)
Data	Seeded, realistic FIR dataset covering 8 districts, 4 crime categories, 400+ case records
Tooling	Git / GitHub (PR-based workflow) · Zoho Catalyst CLI

4. Proposed Impact & Use Case

- **Command staff & analysts** — instantly see where and how crime is trending across districts and categories, to direct patrols and resources where they are needed most.
- **Investigating officers** — a single workspace per case surfaces every linked entity, the timeline, evidence and AI-flagged patterns, cutting the time spent piecing together a case file by hand.
- **The department** — a low-cost, serverless, scalable foundation built entirely on Zoho Catalyst's pay-as-you-go cloud, extensible with more data sources, alerting, evidence file storage and scheduled reporting without re-architecting.
- **Citizens (future extension)** — the same crime-statistics views can be safely surfaced publicly for transparency, in the same dignified government-portal style used throughout the app.

Raksha-Drishti — "protection through vision." A public-safety utility of the State Government.